ANKIT RAJ

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SUMMARY

Software engineer with hands-on experience designing REST APIs, relational and NoSQL data models, and cloud-deployed backend systems. Strong foundation in Object-Oriented Programming, Data Structures & Algorithms, and system design. Fast learner focused on writing scalable, maintainable, production-quality code.

SKILLS

- **Languages**: Java, Python, JavaScript (ES6+), C, C++, SQL
- **Backend**: Spring Boot, Flask, Node.js, Express.js, REST APIs, JWT Authentication, Maven, Redis
- **Framework**: React.js, Next.js, LangChain, LangGraph, HTML, CSS
- **Databases**: PostgreSQL, MySQL, MongoDB
- **Tools**: Git, GitHub, Postman, GCP, Excel
- **Core CS & Concepts**: Data Structures & Algorithms, OOP, System Design, Collections, Exception Handling, Multithreading, XML/JSON

EXPERIENCE

Data Automation Engineer Intern — July 2025 - Present

Toyam Investigation and Valuation Pvt. Ltd — Hybrid

- Developed a Chrome browser extension using **JavaScript** and **Manifest V3** to automate **web data scraping**, collecting 120+ records and eliminating repetitive manual data entry.
- Built an end-to-end document automation pipeline using **Google Cloud Vision OCR**, **Retrieval-Augmented Generation (RAG)** with **LangChain** and **Chroma**, and the **Gemini API** to extract and process unstructured documents, automating 70-80% of manual work.
- Engineered a WhatsApp automation bot integrated with the Toyam app that downloads exported chat data, parses it into structured form, and uploads it to Google Drive or packages it as ZIP archives, processing 250+ exports.
- Unified scraping, parsing, and cloud storage into a single automated workflow, cutting manual handling and reducing human error by 30%.

Software Engineer Intern — April 2025 - June 2025

HackerRank — Remote

- Led a team of four to develop a high-performance **SQL** server using **Python**, **Flask**, and **SQLite3**.
- Built **REST APIs** for efficient data handling and seamless client-server communication.
- Implemented efficient **database management** and **query handling**, ensuring reliability and speed.
- Adopted Agile workflow for collaborative problem-solving and iterative development.
- Collaborated with cross-functional teams to align development with business requirements.

PROJECTS

GeoPharma: Smart Medical Assistance Platform — May 2025

Tools: *React, Tailwind CSS, EasyOCR, ExpressJS, Rest API*

- Developed a dynamic medical website enabling prescription reading and medicine suggestions through a user-friendly interface.
- Created **REST APIs** with **EasyOCR** library for text extraction, integrated medicine info, alternatives, and buy links via pharmacy APIs.
- Deployed frontend on **Vercel** and backend on **Render** with responsive UI, accessibility, and smooth performance.

Aria Voice Agent: Voice-Based Customer Support Assistant — October 2026

Tools: *Python, Flask, Web Speech API, REST APIs, Vercel, Render*

- Built a real-time voice agent using **Web Speech API** for speech recognition and synthesis, handling 12 customer-support intents with no paid APIs.
- Engineered a rule-based dialogue manager with **regex** intent and entity extraction, replying in under 2 ms per turn.
- Designed a stateless architecture with client-side call context for multi-turn follow-ups, deployed on **Vercel** and **Render**.
- Handled unclear audio, invalid order IDs and Hinglish keywords; verified across 24 scripted conversation scenarios.

EDUCATION

Amity University Jharkhand — May 2023 – May 2027

Bachelor of Technology, Computer Science and Engineering — CGPA: 7.9

Bokaro Ispat Senior Secondary School, CBSE

Class XII (Senior Secondary): 72% — March 2023

MGM Higher Secondary School, CBSE

Class X (Secondary): 92% — March 2021

ACHIEVEMENTS

- **Elite Coder Hackathon Winner (2025)**: Developed a full-stack investment platform enabling women to manage portfolios using customized analytics tools
- **IEEE Paper Presentation (2025)**: Research paper accepted and presented at an IEEE international conference, contributing to peer-reviewed technical knowledge

CERTIFICATIONS

- Node.js: Backend Development Bootcamp, Udemy: — June 2025
  - Mastered backend development with Node.js, Express, and REST APIs via Udemy.
- MERN Stack: All You Need to Know with Practical Project, Udemy: — June 2025
  - Mastered React components and rendering optimizations to make UI load and respond multiple times faster.